$P(X = x) = nC_x\, p^x\, q^{n-x}$, x= 0,1,2,...n

It is a Binomial Distribution with n=4

$p = \frac{80}{100} = \frac{4}{5}$ $\quad q = 1\text{-}p = 1 - \frac{4}{5} = \frac{1}{5}$

$$P(x = 0) = 4C_0 (\tfrac{4}{5})^0 (\tfrac{1}{5})^4 = (\tfrac{1}{625})$$

$$P(x=1) = 4C_1 (\tfrac{4}{5})(\tfrac{1}{5})^3 = \tfrac{16}{625}$$

$$P(x=2) = 4C_2 (\tfrac{4}{5})^2(\tfrac{1}{5})^2 = (\tfrac{96}{625})$$

$$P(x = 3) = 4C_3 (\tfrac{4}{5})^3(\tfrac{1}{5})^1 = (\tfrac{256}{625})$$

$$P(x = 4) = 4C_4 \left(\tfrac{4}{5}\right)^4 \left(\tfrac{1}{5}\right)^0 = \left(\tfrac{256}{625}\right)$$

The probability distribution is given below.

$X=x$	0	1	2	3	4
$P(X=x)$	$\frac{1}{625}$	$\frac{16}{625}$	$\frac{96}{625}$	$\frac{256}{625}$	$\frac{256}{625}$

Model values are 3 and 4 (Maximum number of occurrences)

Example 10.10

In a college, 60% of the students are boys. A sample of 4 students of the college, is taken, find the minimum number of boys should it have so that probability up to that number is $\geq \frac{1}{2}$.

Solution:

It is given that 60% of the students of the college are boys and the selection probability for a boy is 60% or 0.6 As we are taking four samples, the number of trials n = 4. The selection process is independent.

$X \sim B\ (n, p)$

$P\ (X = x) = \quad nC_x\, p^x\, q^{n-x}$, x= 0,1,2,...n

Let X be the number of boys so that $P(X \leq x) \geq \frac{1}{2}$

If $x = 0$ $\quad P(X \leq 0) = 4C_0 (\tfrac{3}{5})^0 (\tfrac{2}{5})^4 = \tfrac{16}{625} < \tfrac{1}{2}$